KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

COLLEGE OF ENGINEERING

B.Sc. (Engineering), Mid Semester Examination, November, 2019
Third Year

TE 385: Signals and Systems ***TIME: 1 hour***

Program of study: TE/BME INDEX NO:

Attempt all questions

SECTION A (10 Marks)

Fill in the blanks

1 Signals that have finite signal energy are referred to as ....Energy signals....

2 Every signal can be decomposed into even and odd parts. True/False

3 A shift of a signal to the right is a ....delay....

4 Fourier series applies to only ....periodic....signals.

5 Discrete-time signal may be obtained by ....sampling.... ... continuous-time signal

6 The convolution of a signal with an impulse is equal to ....the orignal signal....

7 $t\delta(t) =$ ....0...., for all values of t.

8 If $x(t)$ is replaced by $x(at)$, then $x(at)$ is ....a linearly compressed.... version of $x(t)$ if $|a| > 1$

9 Any discrete-time signal can be expressed as ....a sequence.... of time-shifted impulses

10 For an even signal $x(-t) =$ ....x(t).... for all t.

Scanned with CamScanner

SECTION B (50 Marks)

1 What is a periodic signal? Determine whether or not each of the following signals is periodic. In case a signal is periodic, specify its fundamental period.

(a) $x(t) = 10\cos\left(50\pi t + \frac{\pi}{6}\right)$

(b) $x[n] = 4\cos\left(4\pi n + \frac{\pi}{6}\right)$

2 Consider the system $\mathbf{y(n) = \sin x[n]}$, state whether the system is

(a) Memoryless
(b) Linear
(c) Time invariant
(d) Causal

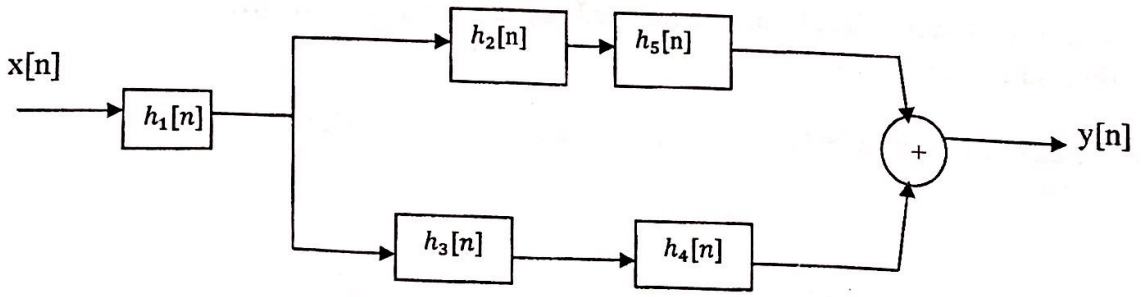
Justify your answers

3 Compute the convolution of x(t) = u(t) and h(t) = u(t)

4 (a) If $y[n] = x[n] * h[n]$, show that $\Sigma_y = \Sigma_x \Sigma_h$ where $\Sigma_x = \Sigma_{n=-\infty}^{\infty} x[n]$ and $\Sigma_h = \Sigma_{n=-\infty}^{\infty} h[n]$.

(b) Compute $y[n] = x[n] * h[n]$ when x[n] = {3, 1, 3} and $h[n] = x[n]$ and check the correctness by using (a) above.

5 Consider the interconnection of LTI systems as shown below. Express the overall impulse response in terms of $h_1[n], h_2[n], h_3[n]$, $h_4[n]$ and $h_5[n]$



Scanned with CamScanner

6 Convert the function $g(t) = (1+j)e^{j4\pi t} + (1-j)e^{-j4\pi t}$ to an equivalent form in which j does not appear.

7 Consider the signal $x[n] = \{0, 0, 1, \underset{\uparrow}{1}, 2, 2, 1, 1\}$

Sketch and label carefully the following: (a) $x[n+1]$ b) $x[1-n]$ c) $x[n]u[-n]$

d) $x[n-1]\delta[n-2]$ e) $x[n^2]$

K. A. P. Agyekum

Scanned with CamScanner